

A candidate human surface-protein universe, *merged from five independent sources.*

The Deliverome Project · A reference universe for transparent cross-source comparison · 2026-04-17 (revised 2026-05-05)

ABSTRACT

Selecting therapeutic-delivery targets on the human cell surface requires an unbiased starting list – one that captures every credible claim of surface expression without prejudging which evidence type is most trustworthy. We merge five independent annotation sources – UniProt, Gene Ontology, SURFY (ML classifier), CSPA (mass spec), and Human Protein Atlas v25 (immunofluorescence). Each source contributes one independent surface-call under its own rule, harmonized to current UniProt primary accessions. DeepTMHMM topology predictions (partial cohort) and JensenLab COMPARTMENTS calls (corroboration-gated, contributes no unique members) are attached per row as auxiliary evidence but are omitted from universe-level figures. The resulting universe contains **5,680 proteins** – **188** supported by all five, **659** by at least four. Most sit in lower agreement classes – an expected pattern preserved for downstream prioritization.

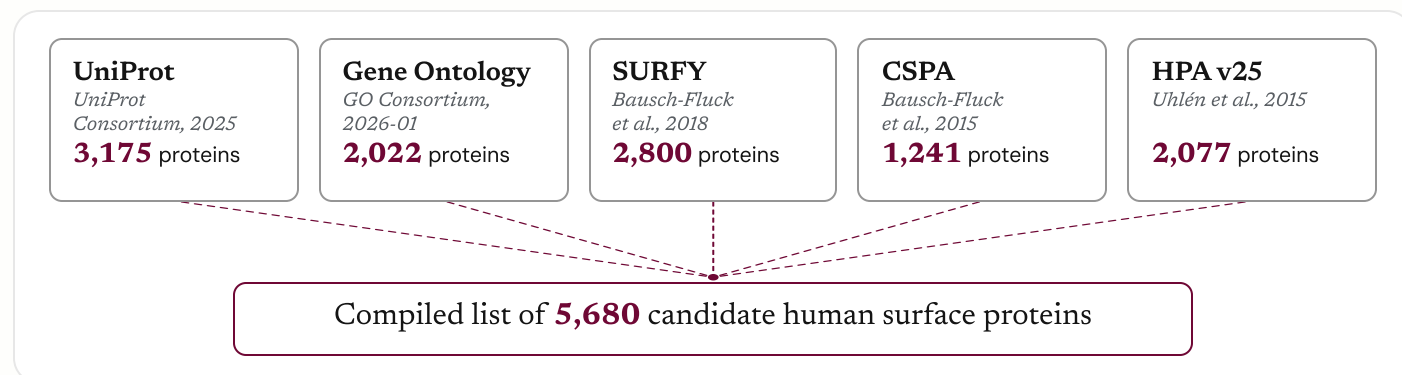


FIGURE 1 Five independent sources of human surface-protein evidence are reconciled against current UniProt primary accessions and merged into a single candidate universe. Per-source counts are after each source's surface-evidence rule is applied. A protein enters the universe if at least one source flags it as surface. DeepTMHMM topology calls (partial cohort, § 2.5) and JensenLab COMPARTMENTS calls (corroboration-gated, contributes no unique members; § 2.7) are attached per row as auxiliary evidence but are omitted from the universe-level figures.

HEADLINE · 5,680 CANDIDATE PROTEINS

188 supported by all 5 sources · 659 by at least 4

The tightest pairwise agreement is UniProt with SURFY (Jaccard 0.67) – both deeply curated for plasma-membrane / surface localization. DeepTMHMM (partial cohort) and JensenLab COMPARTMENTS (corroboration-gated, contributes no unique members) are attached as per-row auxiliary evidence but are omitted from universe-level figures.

§ 1 Data sources

#	SOURCE	SNAPSHOT	EVIDENCE MODALITY	ROWS	SUPPORTED
1	UniProt human surface-candidate query (subcellular location + topology)	Release 2026_01	Curated database	3,175	3,175
2	Gene Ontology annotations (three cellular-component roots – see § 2.2)	2026-01	Curated database	2,557	2,022
3	SURFY (Bausch-Fluck et al., 2018)	2018	ML classifier	2,886	2,800
4	CSPA (Bausch-Fluck et al., 2015)	2015	Mass-spec surfaceome	1,500	1,241
5	Human Protein Atlas (subcellular localization)	v25 / 2026-01	Antibody immunofluorescence	3,004	2,077



§ 2 Methods

All seven sources (five gating, two auxiliary) are harmonized against UniProt release 2026_01³ before the merge. For each source we describe what it measures, the rule under which a protein is counted, and the main reason that source disagrees with the others.

§ 2.1 UniProt 3,175 SUPPORTED

WHAT IT MEASURES The Swiss-Prot reviewed human proteome¹, filtered to entries whose curator-assigned subcellular-location comments and topology features² place the protein on the cell surface.

INCLUDED An entry is UniProt-supported if either (a) its subcellular- location comment contains one of the strict surface terms "Cell surface", "Apical cell membrane", "Basolateral cell membrane", or "GPI-anchor", or (b) the entry carries at least one topological-domain feature annotated Extracellular. Apical and basolateral name the two faces of the plasma membrane; an extracellular topology feature confirms a residue stretch sits outside the cell.

EXCLUDED The broad "Cell membrane" subcellular-location comment is *not* counted on its own – it conflates extracellular-, lateral-, and cytoplasmic-face plasma-membrane proteins, including peripheral proteins anchored on the inner leaflet that aren't accessible on intact cells. The upstream UniProt fetch query still pulls these so the cached snapshot supports a future broader sweep, but the M1 surface flag is set only via the two strict clauses above. Unreviewed (TrEMBL) entries and entries whose subcellular location is exclusively internal-organelle without an extracellular topology likewise do not enter the pool.

WHY IT DISAGREES Evidence depth varies by gene: well-studied proteins carry direct experimental annotations; less-studied ones rely on orthologue transfer. The Extracellular clause also admits some multi-location proteins; we retain them deliberately.

§ 2.2 Gene Ontology 2,022 SUPPORTED

WHAT IT MEASURES Curated gene-term associations for subcellular location⁴. Each association carries an evidence code – experimental, curated, sequence-inferred, or purely electronic.^{5,6}

INCLUDED Three cellular-component roots plus their `is_a` + `part_of` descendants: **cell surface** (GO:0009986), **external side of plasma membrane** (GO:0009897), and **integral component of plasma membrane** (GO:0005887). Counted as GO-supported if the protein has at least one association to an included root (or descendant) backed by an experimental, curated, or non-electronic sequence-inference code.

EXCLUDED The parent **plasma membrane** (GO:0005886) is excluded as too broad – it admits cytosolic-facing PM-associated proteins; its PM-specific members are already covered by the children. The MF terms **signaling receptor activity** (GO:0038023) and **TM signaling receptor activity** (GO:0004888) are excluded because they capture *activity* regardless of location and admit endosomal TLR3/7/8. Anchored-component terms (GO:0031225, GO:0046658) are obsolete in current GO; GPI/lipid-anchor evidence comes via UniProt `ft_lipid`. IEA-only annotations are kept as provenance, not counted.

WHY IT DISAGREES Deep for well-studied receptors, thinner for less-curated surface proteins; IEA-only support is excluded.

STRICT-SURFACE RULE

Two clauses OR-ed: (a) any of four `cc_scl_term` keywords (Cell surface, Apical CM, Basolateral CM, GPI-anchor); or (b) `ft_topo_dom:Extracellular` fires. Plain Cell membrane is fetched but not counted.

IN THIS SNAPSHOT

Topology does most of the work: 2,992 of 3,175 supported rows have an Extracellular feature; only ~415 match a strict location keyword.

2026_01

UNIPROT RELEASE

WHY DROP MF TERMS

Admitting GO:0038023 / GO:0004888 pulls in endosomal TLR3/7/8 and other intracellular-membrane signaling receptors on activity annotation alone – incompatible with "accessible on intact cells."

IEA-ONLY

Retained as provenance (`go_low_confidence_only = 1`); never counted.



§ 2 Methods *continued* – SURFY & CSPA

§ 2.3 SURFY 2,800 SUPPORTED

WHAT IT MEASURES A random-forest classifier trained by Bausch-Fluck et al.⁷ on a curated gold-standard surfaceome using sequence, topology, and annotation features. Assigns every human protein a surface / not-surface label and a confidence score.

INCLUDED Counted as SURFY-supported if the protein's published SURFY label is "surface". We use the label as provided by the authors; no additional score cutoff is re-applied.

EXCLUDED Proteins labeled "nonsurface" and entries left unclassified in the SurfaceomeMasterTable are excluded and dropped from the merge input (rather than carried as provenance) because SURFY covers the full reviewed proteome – keeping them would only inflate the zero-support bucket without adding information.

WHY IT DISAGREES SURFY tends to over-call. A protein with a transmembrane domain and sequence hallmarks of membrane insertion tends to score positive whether or not it actually reaches the plasma membrane.

§ 2.4 CSPA 1,241 SUPPORTED

WHAT IT MEASURES The Cell Surface Protein Atlas (Bausch-Fluck et al., 2015)⁸: a mass-spectrometry surfaceome built by chemically labeling N-linked glycopeptides on the surface of intact, living cells across many human cell lines. Each detection is categorized in CSPA Table_B as **1 – high confidence**, **2 – putative**, **3 – unspecific**, or blank (detected in Table_A but not assigned a Table_B class).

INCLUDED Counted as CSPA-supported if any pre-accession-history-collapse row has category **"1 – high confidence"** or **"2 – putative"**. The flag is a boolean OR across pre-collapse rows, so a primary that inherited evidence from multiple merged historical accessions (e.g. HLA alleles) gains the flag if any contributing row was high-confidence or putative.

EXCLUDED **"3 – unspecific"** detections (non-specific captures / contaminants) and **blank-category** rows (detected but not classified) are retained in the merge as provenance (`cspa_surface_flag = 0`) so downstream can distinguish "absent from CSPA" from "seen only as unspecific" from "detected but not classified." They do not contribute to the positive flag.

WHY IT DISAGREES CSPA only observes proteins expressed on the specific cell lines sampled. Absence is not evidence of non-surface status – only absence from the sampled contexts.

KNOWN FAILURE

ABCB9, a lysosomal ABC transporter, is flagged by SURFY despite residing on an internal membrane. Strong disagreement between SURFY and the experimental sources (CSPA, HPA) is itself an informative signal.

SCOPE CAVEAT

Low pairwise agreement between CSPA and the database-driven sources (0.08–0.32) reflects this sampling scope, not data quality.



§ 2 Methods *continued* – DeepTMHMM & Human Protein Atlas

§ 2.5 DeepTMHMM PER-ROW EVIDENCE · PARTIAL COHORT

WHAT IT MEASURES A deep-learning transmembrane-topology predictor⁹. Assigns one of five labels per sequence: TM (alpha-helical transmembrane), SP+TM (signal peptide + TM helices), BETA (transmembrane beta-barrel), SP (signal peptide only – predicted secreted), or GLOB (globular, no TM or SP).

ROLE IN M1 DeepTMHMM is treated strictly as **per-row auxiliary evidence**. It does not contribute to universe membership (a topology call says "this sequence is membrane-embedded," not "this protein is on the cell surface" – the predictor cannot distinguish the plasma membrane from ER, Golgi, lysosomal, or mitochondrial-outer membranes), and it is omitted from the universe-level figures and analysis in this milestone because it was run on only 2,360 of the reviewed-proteome accessions. Including a partial cohort in agreement counts and Jaccard heatmaps would conflate scope gaps with disagreement.

SURFACE LABEL The `deeptmhmm_surface_flag` is emitted on every universe row (1 when any run on the canonical sequence or isoforms returns TM or SP+TM; 0 otherwise). BETA is excluded because in humans these predictions overwhelmingly identify mitochondrial outer-membrane proteins (VDAC1/2/3, TOMM40, SAMM50). SP and GLOB are excluded by definition (secreted, non-membrane). The raw label is preserved on every row so downstream agent assessment can re-inspect the topology call directly.

RE-INCLUSION CRITERION DeepTMHMM will be added back to the universe-level figures once it has been run on the full reviewed human proteome.

§ 2.6 Human Protein Atlas v25 2,077 SUPPORTED

WHAT IT MEASURES An antibody-based immunofluorescence atlas of subcellular localization.^{10,11} Each gene is scored into named localizations (plasma membrane, cell junctions, vesicles, endosomes, lysosomes, Golgi, ER, nucleus, cytosol, mitochondrion, ...) with a per-localization reliability tier: **Enhanced**, **Supported**, **Approved**, or **Uncertain**.¹²

INCLUDED Counted as HPA-supported if either **Plasma membrane** or **Cell Junctions** appears at **Enhanced**, **Supported**, or **Approved** in the per-localization tier column (not gene-wide Reliability). Cell junctions count: E-cadherin, claudins, JAM-family, occludin, and desmosomal cadherins carry genuine extracellular-accessible domains on intact epithelia.

EXCLUDED PM or Cell Junctions at **Uncertain** reliability are excluded. HPA's **Extracellular-location** column is excluded entirely: in v25 it is populated by "Predicted to be secreted" – a SignalP-based sequence prediction, not an imaging observation – so counting it would double-count a predictor signal and conflate secreted with surface-accessible. **Vesicle, endosome, and lysosome** staining is excluded from the flag because cycling between internal compartments and the surface is not evidence of steady-state surface residence; it is carried as provenance (`hpa_trafficking_associated`). Nuclear, cytosolic, Golgi, ER, and mitochondrial localizations are excluded as not surface-relevant.

WHY IT DISAGREES HPA coverage is antibody-dependent. Proteins without a validated antibody at the v25 release, or whose surface localization is limited to cell types not sampled, will be absent.

WHY DROP BETA

In humans, beta-barrel predictions overwhelmingly identify mitochondrial outer membrane proteins (VDAC1/2/3, TOMM40, SAMM50) – not plasma-membrane / cell-surface proteins.

SCOPE

2,360 of ~20k reviewed human proteins have been processed. Extending DeepTMHMM to the full proteome is a planned next step; figures will be re-generated with DeepTMHMM included once that completes.

WHY PER-TIER

Using the *per-localization* tier (not the gene-wide Reliability) avoids overcalls where a gene's strongest signal is nuclear/cytosolic while its PM call lands in Uncertain.

TRAFFICKING ≠ SURFACE

Vesicle/endosome/lysosome staining carried as provenance only. Admitting it on its own would reintroduce ABCB9-class internal-membrane false positives.

§ 2 Methods *continued* – JensenLab COMPARTMENTS

§ 2.7 JensenLab COMPARTMENTS AUXILIARY PER-ROW EVIDENCE ·

CORROBORATION-GATED

COMPARTMENTS is *attached* to each row of the candidate-universe TSV but does not gate universe membership or contribute to the *k*-of-5 agreement score. By design, the corroboration gate (below) requires a COMPARTMENTS-supported protein to also be flagged by another source – so COMPARTMENTS contributes **zero unique members** to the universe (verified: 148 of 148 COMPARTMENTS-flagged proteins are also flagged by ≥1 of UniProt/GO/SURFY/CSPA/HPA). Its value is purely as a confidence-boosting per-row vote for downstream consumers.

WHAT IT MEASURES A meta-database (Binder et al., 2014)¹³ integrating four evidence streams per (protein, location): a **knowledge** channel from curator-assigned annotations, an **experiments** channel primarily ingesting immunofluorescence, a **text-mining** channel from automated co-mention in biomedical literature, and a **predictions** channel from sequence-based localization predictors (WoLF PSORT + YLoc-HighRes). Scores are reported on a 0–5 star scale per channel per term.

INCLUDED The **text-mining** channel and the non-HPA portion of the **experiments** channel are used as independent evidence¹⁴. Counted as COMPARTMENTS-supported when both hold: (a) $\max(\text{experiments}, \text{textmining}) \geq 3$ stars across the strict surface GO-term set {G0:0009986, G0:0009897, G0:0005887} – kept in lockstep with § 2.2's GO inclusion set, and (b) at least one of the other DB sources independently flags the protein as surface under its own rule (`compartments_corroborated == 1`).

EXCLUDED The broad parent terms G0:0005886 (plasma membrane) and G0:0031225 (anchored component of *any* membrane) are dropped to mirror § 2.2: both pull in cytoplasmic-face PM proteins or non-PM-anchored proteins, and their PM-specific members are already covered by the children. The **knowledge** channel is excluded because it re-ingests GO and UniProt-SubCell – both already first-class sources here; counting it would triple-count curator evidence. The **predictions** channel is excluded because it wraps WoLF PSORT + YLoc-HighRes – sequence-based predictors in the same family as SURFY and DeepTMHMM; empirically it drives ~73% of pre-filter hits and is ~100% PSORT-on-G0:0005886. Within the experiments channel, rows whose upstream source is "HPA" are dropped before scoring to avoid double-counting the first-class HPA IF evidence. All four channels' per-term stars are retained as provenance columns.

WHY IT IS HELD OUT OF THE FIGURES By construction COMPARTMENTS cannot admit a protein on its own – and against the live build, it admits **zero** proteins that another source has not already admitted. Including it as a sixth gating vote would therefore inflate the agreement score for already-corroborated proteins on ontology grounds the GO source itself rejects (text-mining co-occurrence with G0:0005886), without ever expanding the universe. We hold it out of the universe-level figures and treat it as a per-row confidence cue.

WHY CORROBORATION

Text-mining alone picks up legitimate co-occurrences with "plasma membrane" for transcription factors (TP53, MYC), cytokines (IL1B, IFNG, IL4), and secreted serum proteins (ALB, APOE, CRP) that are not surface-accessible on intact cells.

§ 2.8 Source dependencies

The seven sources (five gating, two auxiliary) do not sit side by side – several ingest each other, and a handful of external upstream tools feed specific primaries. These edges matter when reading the k-of-5 agreement on the next page: apparently independent sources often share upstream inputs, and the auxiliary aggregator (COMPARTMENTS) explicitly integrates evidence from several others – which is one reason it is held out of the gating set.

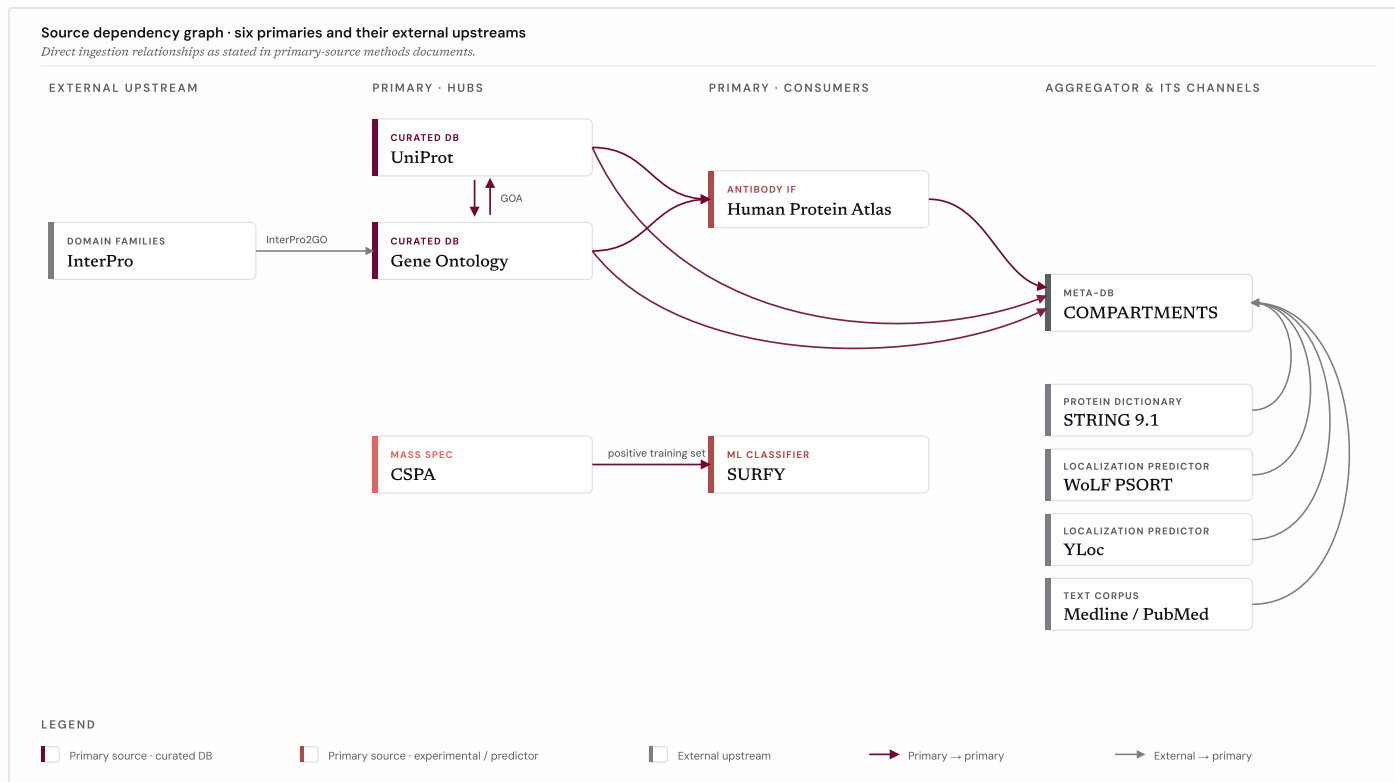
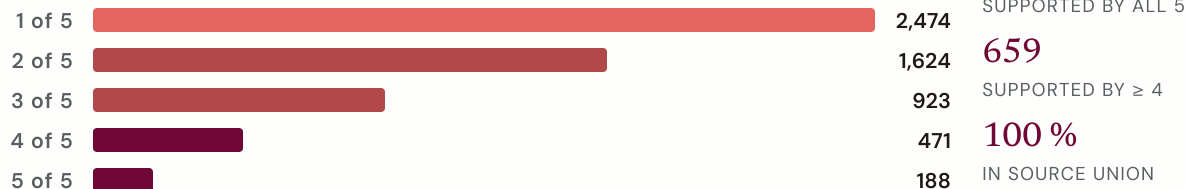


FIGURE 2 Source dependency graph. Direct ingestion relationships as stated in the primary-source methods documents linked in §§ 2.1–2.7. UniProt ↔ Gene Ontology form a bidirectional hub via the GOA pipeline (*Keyword2GO*, *SubcellularLocation2GO*, *InterPro2GO*). CSPA feeds SURFY's positive training set. HPA consults UniProt for knowledge-based annotation and emits GO cellular-component identifiers. COMPARTMENTS is the main aggregation sink – integrating UniProt comments, GO term mapping with ontology propagation, HPA imaging data, STRING 9.1's protein dictionary, WoLF PSORT and YLoc predictions, and Medline abstracts. DeepTMHMM (currently a partial-cohort per-row evidence track; § 2.5) is omitted from the graph until the full proteome has been processed.

§ 3 Results

The agreement distribution and pairwise Jaccard overlaps quantify where the five gating sources converge, and – more usefully – where they don't. Differences across sources are expected and help define where the evidence is strongest or still uncertain.

§ 3.1 k-of-5 agreement distribution



One hundred eighty-eight proteins are supported by all five sources; another **471** by four – together **659** ($\approx 11.6\%$ of the universe) form the high-confidence core. The 1-of-5 tail is dominated by proteins that only the UniProt curator-assigned query catches. A further **1,704** accessions appear in a raw source but fail every positive-flag rule (GO-IEA-only, CSPA unspecific/blank, HPA secreted-only, split-ambiguous remaps); those are emitted to `candidate_universe_zero_support.tsv` and are not part of the main universe.

The 5-of-5 and 4-of-5 classes define the high-confidence core. The middle classes (2-of-5 and 3-of-5) are often the most biologically interesting – disagreement frequently carries real meaning (predictor over-calls, cell-line sampling gaps, missing HPA antibodies).

§ 3.2 Pairwise agreement

UniProt $\cap$ SURFY curator + ML classifier	0.669
UniProt $\cap$ Gene Ontology	0.293
SURFY $\cap$ CSPA	0.287
Gene Ontology $\cap$ SURFY	0.261
UniProt $\cap$ CSPA	0.223
Gene Ontology $\cap$ CSPA	0.178
UniProt $\cap$ HPA	0.166
Gene Ontology $\cap$ HPA	0.161
SURFY $\cap$ HPA	0.157
CSPA $\cap$ HPA	0.118

UniProt $\cap$ SURFY (0.67) leads – both are deeply curated for plasma-membrane / surface localization, and the strict-surface pruning of UniProt's inclusion rule (§ 2.1) tightened that overlap. GO overlaps (0.26–0.29) reflect its cellular-component- only scope after MF receptor-activity terms were excluded. CSPA and HPA pair lowest with each other (0.12), reflecting CSPA's cell-line-restricted sampling and HPA's antibody-dependent coverage rather than data quality.

188

SUPPORTED BY ALL 5

659

SUPPORTED BY ≥ 4

100 %

IN SOURCE UNION

INVARIANT

By construction, every protein in the universe is supported by at least one of the five gating sources.

COMPARTMENTS (auxiliary, corroboration-gated; § 2.7) and DeepTMHMM (auxiliary, partial cohort; § 2.6) are attached per row but do not contribute to this distribution.

WELL-KNOWN EXAMPLES

Canonical receptors and HLA class I/II entries appear in the universe with strong cross-source corroboration where each source has the coverage to weigh in, and fall to lower $k/5$ only on the sources whose coverage is genuinely thin (chiefly HPA's gaps on individual HLA alleles and immunoglobulin class-switch variants). Per-gene $k/5$ attributions will be re-derived against the current build before the LLM reconciliation step keys on them.

§ 3.3 Source-agreement figures

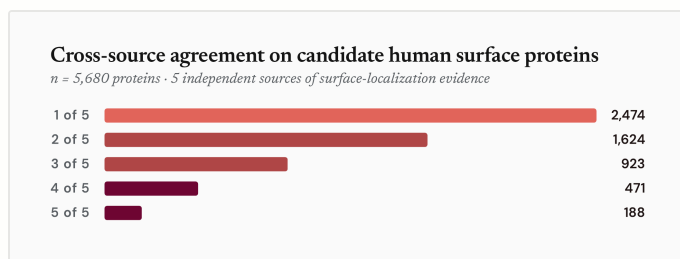


FIGURE 3 k-of-5 agreement. 188 of the 5,680 universe proteins are supported by all five gating sources; 659 by at least four. DeepTMHMM (partial cohort) and COMPARTMENTS (corroboration-gated, contributes no unique members) are omitted from this distribution.

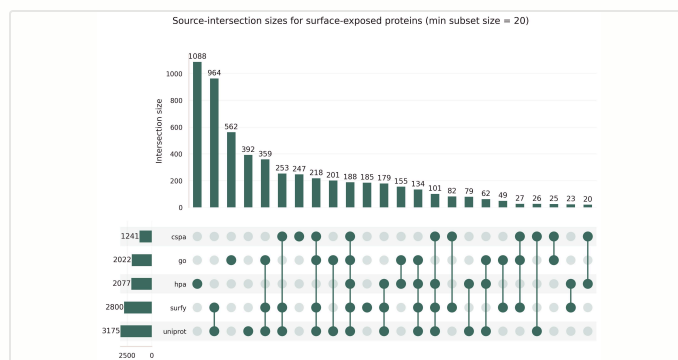


FIGURE 4 Co-occurrence across sources. The most populated subsets are UniProt-only and UniProt $\cap$ SURFY. Rendered from the five gating sources only; DeepTMHMM and COMPARTMENTS are auxiliary in this milestone and are omitted from this view.

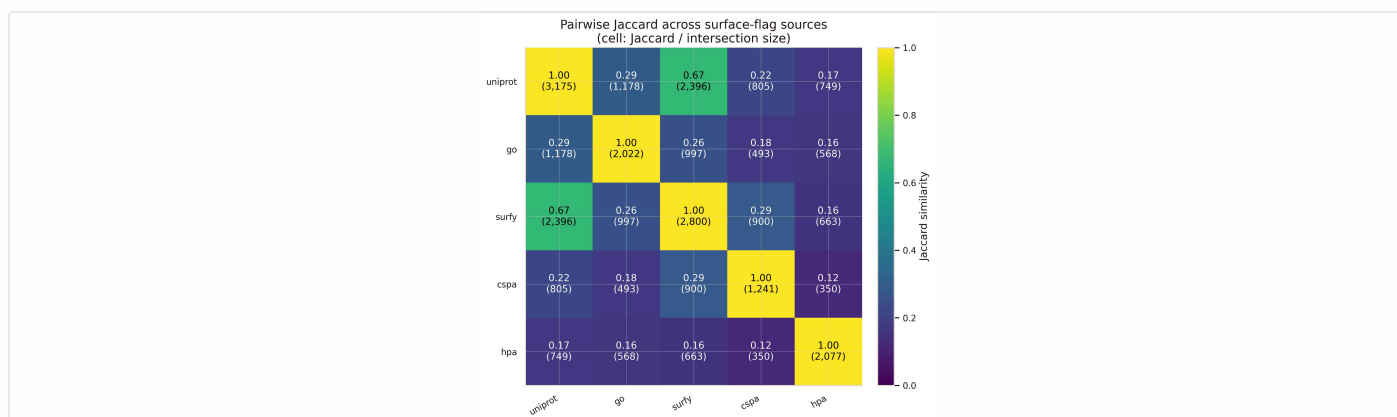


FIGURE 5 Pairwise agreement. UniProt $\cap$ SURFY (0.67) leads. Low values involving CSPA reflect cell-line-restricted sampling, not data quality. Rendered from the five gating sources only.

§ 4 Limitations and next steps

This milestone produces a broad, traceable candidate list for downstream filtering, not a validated surfaceome. Four limitations matter for interpretation.

- DeepTMHMM coverage.** Per-row auxiliary evidence only (2,360-protein cohort); omitted from figures (§§ 2.5, 3). Extending it – with SignalP 6.0 and GPI-anchor predictors – to the full proteome will let us re-include it as a gating source.
- COMPARTMENTS contributes no unique members.** The corroboration gate (§ 2.7) drops COMPARTMENTS-only signals to keep TFs / cytokines / serum proteins out. Verified against the live build: 148 of 148 COMPARTMENTS-flagged proteins are also flagged by ≥ 1 of the five gating sources, so COMPARTMENTS is held out of universe-level figures and treated as a per-row confidence cue.
- Topology-driven UniProt admissions.** $\approx 94\%$ of UniProt entries (2,992 / 3,175) qualify via Extracellular topology rather than a strict keyword (§ 2.1).
- Tissue / cell-type filtering.** The full HPA product (tissue, single-cell, cancer atlas, Secretome, Predicted-membrane) is not yet incorporated.
- Ambiguous gene symbols.** 5 input symbols resolved ambiguously through MyGene; emitted with `gene_symbol_mapping_status = "ambiguous"`.



§ 5 References

First-party citations for each source's methodology and data, grouped by provider. Each entry includes direct view / PDF / download links and a one-line description of what the reference is cited for in §§ 2.1–2.7.

§ 5.1 UNIPROT

- [1] UniProt Consortium. *UniProt: the Universal Protein Knowledgebase in 2025*. Nucleic Acids Research 53(D1):D609–D617, 2025.
view · pdf · UP000005640
Canonical description of UniProtKB architecture, the Swiss-Prot/TrEMBL review split, and the 2025 content update. CITED § 2.1
- [2] UniProt. *Subcellular location* (help page).
view · subcell.txt
Controlled vocabulary used in SUBCELLULAR LOCATION comment lines; the basis for the surface-topology filter (cell surface, integral, GPI-anchored, transmembrane). CITED § 2.1
- [3] UniProt. *Release 2026_01*. Release notes, 2026-01-28.
release notes · downloads
The exact snapshot used for accession-history reconciliation across all sources. Release cadence is ~8 weeks. CITED § 2 INTRO

§ 5.2 GENE ONTOLOGY

- [4] Gene Ontology Consortium. *Download Annotations*. Current release landing page; human GAF release 2026-03-25.
downloads · goa_human.gaf.gz
The human GO annotation snapshot and its release cadence; source of the gene-term associations filtered to surface-relevant terms. CITED § 2.2
- [5] Gene Ontology Consortium. *GO Annotation File (GAF) format 2.2*.
spec
Defines the single-(gene, term) association record carrying an evidence code, which we filter on to drop purely electronic annotations. CITED § 2.2
- [6] Gene Ontology Consortium. *Guide to GO evidence codes*.
view · GOA IEA pipeline
Evidence-code taxonomy (Experimental, Phylogenetic, Computational Analysis, Author Statement, Curator Statement, Automatically-assigned); the basis for excluding IEA-only annotations from the positive call. CITED § 2.2

§ 5.3 SURFY

- [7] Bausch-Fluck, D., Goldmann, U., Müller, S., et al. *The in silico human surfaceome*. PNAS 115(46):E10988–E10997, 2018.
view · pdf · resource
Random-forest classifier trained on CSPA-positive proteins; source of the 2,886-protein prediction set with 1% / 5% / 15% FPR tiers. CITED § 2.3



§ 5 References *continued* – CSPA, DeepTMHMM, HPA, COMPARTMENTS

§ 5.4 CSPA

- [8] Bausch-Fluck, D., Hofmann, A., Bock, T., *et al.* *A Mass Spectrometric-Derived Cell Surface Protein Atlas*. PLOS ONE 10(4):e0121314, 2015.
[view](#) · [pdf](#) · [S2 \(data\)](#)
Cell Surface Capture glycoproteomics across 41 human and 31 mouse cell types; the validated-surfaceome table used for the CSPA-positive input and its high-confidence / putative / unspecific / blank categories. CITED § 2.4

§ 5.5 DEEPTMHMM

- [9] Hallgren, J., Tsirigos, K. D., Pedersen, M. D., *et al.* *DeepTMHMM predicts alpha and beta transmembrane proteins using deep neural networks*. bioRxiv 2022.04.08.487609, 2022.
[view](#) · [pdf](#) · [service](#)
ESM-1b encoder plus CRF decoder producing the Alpha TM / SP+ TM / Beta / SP-only / Globular class labels used as the topology-based surface call. CITED § 2.5

§ 5.6 HUMAN PROTEIN ATLAS

- [10] Thul, P. J., Åkesson, L., Wiking, M., *et al.* *A subcellular map of the human proteome*. Science 356:eaal3321, 2017.
[view](#)
The original immunofluorescence-based subcellular atlas underlying the HPA subcellular data product. CITED § 2.6
- [11] Human Protein Atlas. *Subcellular localization method – imaging*. Method page.
[view](#) · [license](#)
Current description of antibody validation, reliability scoring, and the knowledge-based comparison against UniProtKB/Swiss-Prot. CITED § 2.6
- [12] Human Protein Atlas. *Subcellular location data (v25.0, Ensembl 109)*. Released 2025-11-11.
[view](#) · [subcellular_location.tsv.zip](#) · [releases](#)
Reliability classes (Enhanced / Supported / Approved / Uncertain), named localizations, and the GO cellular-component identifier column used for the v25 HPA call. CITED § 2.6

§ 5.7 JENSENLAB COMPARTMENTS

- [13] Binder, J. X., Pletscher-Frankild, S., Tsafou, K., *et al.* *COMPARTMENTS: unification and visualization of protein subcellular localization evidence*. Database (Oxford) 2014:bau012, 2014.
[view](#) · [pdf](#)
Defines the knowledge / experiments / text-mining / predictions four-channel integration and the five-star scoring used for the ≥ 3 -star gate. CITED § 2.7
- [14] JensenLab. *COMPARTMENTS – Downloads*. Per-channel TSVs, weekly updated.
[downloads](#) · [textmining.tsv](#) · [experiments.tsv](#)
Per-channel files used to reconstruct the corroboration-gated call (text-mining + non-HPA portion of experiments), holding UniProt / GO / HPA duplicates out of the COMPARTMENTS evidence. CITED § 2.7